A LOWER BOUND FOR SETS LACKING THE CONFIGURATION $\{x, x+y, x+y^3\}$

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Abstract. We give the first explicit lower bound for sets of integers lacking the configuration $\{x, x+y, x+y^3\}$ that goes beyond what sets without cube differences provide: for every $\varepsilon > 0$ there is $A \subseteq \{1, \dots, N\}$ with no such configuration and

$$|A| \gg_\varepsilon N^{\gamma_3 - \varepsilon}, \qquad \gamma_3 = \frac{2}{3} + \frac{3\log 13 + \log 28}{12 \log 91} = 0.8703797389\dots,$$

whereas the best exponent for cube-difference-free sets is $0.8617631\dots$. The construction instantiates Younis's iterated-difference framework with a two-element chain of subsets of $\mathbb{Z}/91\mathbb{Z}$ of sizes $(13, 28)$, the first of which is deliberately smaller than the largest cube-difference-free subset of $\mathbb{Z}/91\mathbb{Z}$. We show that the optimisation over all admissible chains modulo a fixed m is a finite dynamic program over $2^{|P_m|/2}$ states, where P_m is the set of nonzero cubes, and by solving it exactly we prove that no chain of any length modulo 91 gives a larger exponent. A dependency-free verification script is included.

1. Introduction

For a polynomial $f \in \mathbb{Z}[x]$ and an integer $k \geq 2$, two families of configurations in $[N] := \{1, \dots, N\}$ have received particular attention: pairs $\{x, x+f(y)\}$ with $y \neq 0$ (*f-differences*) and triples

$$\{x,\ x+y,\ x+y^k\}, \qquad y \neq 0 \tag{1}$$

(*nonlinear progressions*). Sets of positive density contain both, by the theorems of Sárközy and Furstenberg [4, 12] in the first case (for $f = x^2$; Kamae and Mendès France [5] for general intersective f) and by the polynomial Szemerédi theorem of Bergelson and Leibman [3] in the second. Effective upper bounds for (1) with $k = 2$ are due to Peluse and Prendiville [9, 10], and Peluse [8] handles configurations with polynomials of distinct degrees in finite fields.

In the other direction the only known polynomial-size constructions descend from Ruzsa's base-m construction [11] for square differences, as extended by Younis [13] to k-th power differences, to polynomial differences f with 0 as their only root modulo m, and to the configurations (1). Younis's Theorem 1.5 produces from any *admissible chain* $(R_n)_{n\geq 0}$ of subsets of $\mathbb{Z}/m\mathbb{Z}$ (see Definition 2.1) a set avoiding (1) of size $N^{\gamma-\varepsilon}$ with

$$\gamma = (k-1)\sum_{n\geq 0} \frac{\log_m |R_n|}{k^{n+1}}. \tag{2}$$

For $k = 2$ Younis instantiated this with an alternating pair modulo 65, obtaining $\gamma = 0.7685\dots$ [13, Thm. 1.1]; a longer chain modulo 65 of sizes $(7, 17, 10, 15)$ improves this to $0.7732\dots$ [2]. For $k = 3$ no instantiation appears in the literature: the only bound available for sets lacking $\{x, x+y, x+y^3\}$ is the trivial one obtained from sets lacking cube differences, since $(x+y^3)-x =$

Date: September 2, 2026.

2020 *Mathematics Subject Classification.* Primary 11B25; Secondary 11B30, 05D10.

Key words and phrases. Polynomial Szemerédi theorem, nonlinear Roth theorem, cube differences, difference sets, maximum clique.

y^3. That bound is Ruzsa's exponent $(2+\log_m|R|)/3$ for a cube-difference-free $R \subseteq \mathbb{Z}/m\mathbb{Z}$; the parameters $m = 91$, $|R| = 14$ found by Lewko [6] (see [13, §1]) give $0.8616817797\ldots$, and a recent 116-element set modulo 3367 [1] raises this to $0.8617631316\ldots$. We stress that the two problems are different: a set with no cube differences lacks (1) for $k = 3$, but a set lacking (1) may well contain pairs differing by a cube, and it is exactly this freedom that the construction below exploits. Our result improves the exponent for the configuration problem only; it says nothing about cube differences.

The result.

1.1. **Theorem.** *For every $\varepsilon > 0$ and every positive integer N there exists $A \subseteq [N]$ containing no configuration $\{x, x+y, x+y^3\}$ with $y \neq 0$ such that $|A| \gg_\varepsilon N^{\gamma_3-\varepsilon}$, where*

$$\gamma_3 = \frac{2}{3} + \frac{3\log 13 + \log 28}{12\log 91} = 0.87037973891\ldots.$$

The certificate (Proposition 3.1) is a period-two chain modulo 91 of sizes $(13, 28)$. It has two features worth pointing out. The first set is *not* a largest cube-difference-free set modulo 91 (those have 14 elements): it is the smaller set $\{0\} \cup \{31 + 5t : 0 \le t \le 11\}$, whose 24 nonzero differences have only 10 distinct cubes, and this is what allows the second set to be as large as 28. Second, the value γ_3 is not merely the outcome of a search: Section 4 shows that the optimisation over *all* admissible chains modulo 91, of any length and any structure, is a finite dynamic program with 128 states, and solving it exactly proves that γ_3 is the best exponent Younis's framework can give at this modulus. The same program solves the moduli $7, 13, 19, 31, 37, 43$ exactly and bounds 61 and 67 rigorously, all below γ_3; we do not claim anything about other moduli beyond the searches reported in Section 4.

Conventions. m is always a squarefree positive integer and subsets of $\mathbb{Z}/m\mathbb{Z}$ are identified with subsets of $\{0, 1, \ldots, m-1\}$. For $R \subseteq \mathbb{Z}/m\mathbb{Z}$ and $k \ge 2$ we write

$$\Delta R := \{r - r' \bmod m : r, r' \in R,\ r \neq r'\}, \qquad \Delta R^{(k)} := \{d^k \bmod m : d \in \Delta R\}.$$

Logarithms are natural unless subscripted. All numerical constants quoted in this paper were computed by the scripts distributed with it.[1]

2. Younis's chain construction for k-th powers

2.1. **Definition.** Let $k \ge 2$. A sequence $(R_n)_{n\ge 0}$ of nonempty subsets of $\mathbb{Z}/m\mathbb{Z}$ is *k-admissible* if

$$\Delta R_{n+1} \cap \Delta R_n^{(k)} = \emptyset \qquad \text{for every } n \ge 0, \tag{3}$$

and in addition $\Delta R_1 \cap \Delta R_1^{(k)} = \emptyset$.

The following is Theorem 1.5 of Younis [13]. We include a complete proof because the certificate of Section 3 depends on the exact digit assignment, including the treatment of digit position 0, and because the valuation step deserves to be spelled out.

2.2. **Theorem** (Younis)**.** *Let $k \ge 2$, let m be squarefree, and let $(R_n)_{n\ge 0}$ be k-admissible with $R_0 = \mathbb{Z}/m\mathbb{Z}$. Then for every $\varepsilon > 0$ and every N there is $A \subseteq [N]$ with no configuration* (1) *and $|A| \gg_{m,k,\varepsilon} N^{\gamma-\varepsilon}$, with γ as in* (2)*.*

[1] Code and data accompany the paper as arXiv ancillary files: the verification script of Appendix A, the reproduction script `reproduce.py`, the dynamic program `dpk.py`, the two exact clique solvers `mc.c` and `mis.c`, and the cached transition values.

Proof. For an integer $i \geq 1$ write $k^n \parallel i$ if $k^n \mid i$ and $k^{n+1} \nmid i$. Let $Y \geq 1$ be an integer and let A be the set of integers $\sum_{0 \leq i < Y} u_i m^i$ whose digits satisfy

$$u_i \in \begin{cases} R_0 = \mathbb{Z}/m\mathbb{Z}, & i \geq 1,\ k \nmid i, \\ R_n, & i \geq 1,\ k^n \parallel i\ (n \geq 1), \\ R_1, & i = 0. \end{cases}$$

Suppose $u, v, w \in A$ satisfy $v = u + y$ and $w = u + y^k$ with $y \neq 0$. Then $v \neq u$ and $w \neq u$. Let ℓ be the least index at which the digits of v and u differ and j the least index at which the digits of w and u differ. Then

$$v - u = m^\ell (v_\ell - u_\ell) + \tilde{z}\, m^{\ell+1}, \qquad w - u = m^j (w_j - u_j) + z\, m^{j+1}$$

for some integers $z, \tilde{z}$, where $0 < |v_\ell - u_\ell| \leq m - 1$ and $0 < |w_j - u_j| \leq m - 1$.

We claim $j = k\ell$. Fix a prime $p \mid m$ and write v_p for the p-adic valuation. From the displayed expressions, $\mathrm{v}_p(v - u) \geq \ell$ with equality if and only if $p \nmid (v_\ell - u_\ell)$, and $\mathrm{v}_p(w - u) \geq j$ with equality if and only if $p \nmid (w_j - u_j)$. Since m is squarefree, an integer divisible by every prime factor of m is divisible by m; as $0 < |v_\ell - u_\ell| < m$, some prime $p_1 \mid m$ does not divide $v_\ell - u_\ell$, and similarly some prime $p_0 \mid m$ does not divide $w_j - u_j$. Thus $\mathrm{v}_{p_1}(v - u) = \ell$ and $\mathrm{v}_{p_0}(w - u) = j$. Since $w - u = (v - u)^k$ we have $\mathrm{v}_p(w - u) = k\, \mathrm{v}_p(v - u)$ for every prime p. Taking $p = p_0$ gives $j = k\, \mathrm{v}_{p_0}(v - u) \geq k\ell$; taking $p = p_1$ gives $\mathrm{v}_{p_1}(w - u) = k\ell \geq j$. Hence $j = k\ell$.

Dividing $w - u = (v - u)^k$ by $m^{k\ell} = m^j$ gives

$$(w_j - u_j) + zm = \big((v_\ell - u_\ell) + \tilde{z}m\big)^k, \qquad \text{so} \qquad w_j - u_j \equiv (v_\ell - u_\ell)^k \pmod{m}.$$

Here $w_j - u_j \not\equiv 0 \pmod{m}$ because $0 < |w_j - u_j| < m$, and $(v_\ell - u_\ell)^k \not\equiv 0 \pmod{m}$ because m is squarefree and $0 < |v_\ell - u_\ell| < m$: if $m \mid c^k$ then every prime factor of m divides c, so $m \mid c$.

If $\ell \geq 1$ and $k^n \parallel \ell$ with $n \geq 0$, then $k^{n+1} \parallel j$, so $u_j, w_j \in R_{n+1}$ and $u_\ell, v_\ell \in R_n$, and the congruence exhibits an element of $\Delta R_{n+1} \cap \Delta R_n^{(k)}$, contradicting (3). If $\ell = 0$ then $j = 0$ and all four digits lie in R_1, contradicting $\Delta R_1 \cap \Delta R_1^{(k)} = \emptyset$. Hence A contains no configuration (1).

For the size of A: the number of i with $1 \leq i < Y$ and $k^n \parallel i$ is $\lceil Y/k^n \rceil - \lceil Y/k^{n+1} \rceil \geq (k-1)Y/k^{n+1} - 1$, and for $n = 0$ the number of i with $1 \leq i < Y$ and $k \nmid i$ is $Y - \lceil Y/k \rceil \geq (k-1)Y/k - 1$. Given $\varepsilon > 0$ choose n_0 with $(k-1)\sum_{n \geq n_0} \log_m |R_n|/k^{n+1} \leq \varepsilon$, which is possible since $|R_n| \leq m$. Then

$$|A| \ \geq \ \prod_{n < n_0} |R_n|^{(k-1)Y/k^{n+1} - 1} \ \geq \ m^{-n_0}\, m^{(\gamma - \varepsilon)Y}.$$

Taking $Y = \lfloor \log_m N \rfloor$ gives $\max A < m^Y \leq N$, so $A + 1 \subseteq [N]$, and $|A| \geq m^{-n_0 - \gamma} N^{\gamma - \varepsilon}$. The set $A + 1$ is the required set. □

2.3. **Corollary.** *Let $k \geq 2$, m squarefree, and $A_1, \ldots, A_p \subseteq \mathbb{Z}/m\mathbb{Z}$ nonempty with*

(1) *ΔA_1 contains no nonzero k-th power residue modulo m;*

(2) *$\Delta A_{i+1} \cap \Delta A_i^{(k)} = \emptyset$ for $1 \leq i < p$.*

Then the periodic sequence $(\mathbb{Z}/m\mathbb{Z}, A_1, \ldots, A_p, A_1, \ldots, A_p, \ldots)$ is k-admissible, and Theorem 2.2 applies with

$$\gamma = \frac{k-1}{k} + \frac{k-1}{1 - k^{-p}} \sum_{i=1}^{p} \frac{\log_m |A_i|}{k^{i+1}}.$$

Proof. Condition (3) for $n = 0$ reads $\Delta A_1 \cap \Delta(\mathbb{Z}/m\mathbb{Z})^{(k)} = \emptyset$, and $\Delta(\mathbb{Z}/m\mathbb{Z})^{(k)}$ is the set of nonzero k-th power residues; this is (1). Condition (3) for the pairs (A_i, A_{i+1}), $1 \leq i < p$, is (2). The remaining instances of (3) are the wrap-around pairs (A_p, A_1), and the extra condition

$\Delta A_1 \cap \Delta A_1^{(k)} = \emptyset$; both follow from (1), because $\Delta A_p^{(k)}$ and $\Delta A_1^{(k)}$ consist of nonzero k-th power residues. The set A_i occupies the positions $n \equiv i \pmod p$, $n \geq 1$, in (2), and $\sum_{s\geq 0} k^{-(i+sp)-1} = k^{-i-1}/(1-k^{-p})$. □

3. The certificate for $\{x, x+y, x+y^3\}$

3.1. **Proposition.** *Let* $m = 91$ *and*

$A_1 = \{0, 31, 36, 41, 46, 51, 56, 61, 66, 71, 76, 81, 86\}$,

$A_2 = \{0, 7, 9, 11, 13, 20, 22, 24, 26, 33, 35, 37, 39, 46, 48, 50, 52, 59, 61, 63, 65, 72, 74, 76, 78, 85, 87, 89\}$.

Then $|A_1| = 13$, $|A_2| = 28$, *and conditions (1)–(2) of Corollary 2.3 hold with* $k = 3$, $p = 2$.

Proof. Both conditions are intersections of explicitly computable subsets of $\mathbb{Z}/91\mathbb{Z}$: the 14 nonzero cubes modulo 91 are $\{1, 8, 13, 14, 21, 27, 34, 57, 64, 70, 77, 78, 83, 90\}$, the set ΔA_1 has 24 elements, and the set $\Delta A_1^{(3)}$ of their cubes is $\{1, 8, 14, 27, 34, 57, 64, 77, 83, 90\}$. The dependency-free script in Appendix A performs every check. □

Proof of Theorem 1.1. Apply Corollary 2.3 with $k = 3$, $p = 2$ and the pair of Proposition 3.1:

$$\gamma_3 = \frac{2}{3} + \frac{2}{1-3^{-2}}\Big(\frac{\log_{91} 13}{9} + \frac{\log_{91} 28}{27}\Big) = \frac{2}{3} + \frac{3\log 13 + \log 28}{12 \log 91}. \qquad \square$$

3.2. **Remark** (Structure of the pair)**.** Modulo 13 the set A_2 is the union of the four residue classes $\{0, 7, 9, 11\}$, each taken with all seven residues modulo 7: it is the product of $\mathbb{Z}/7\mathbb{Z}$ with a 4-element subset of $\mathbb{Z}/13\mathbb{Z}$ whose differences $\{2, 4, 6, 7, 9, 11\}$ avoid the nonzero cubes $\{1, 5, 8, 12\}$ of $\mathbb{Z}/13\mathbb{Z}$. The set A_1 is $\{0\}$ together with an arithmetic progression of common difference 5; it meets every residue class modulo 13 exactly once. Its 24 nonzero differences have only 10 distinct cubes because the cube map is 9-to-1 on the 72 units of $\mathbb{Z}/91\mathbb{Z}$ ($91 = 7 \cdot 13$ and both 7 and 13 are $\equiv 1 \pmod 3$) and 3-to-1 on the 18 nonzero non-units, so that the 14 nonzero cubes are the images of 8 classes of 9 units and 6 classes of 3 non-units; the 24 differences of A_1 fall into only 10 of these 14 classes.

3.3. **Remark** (Why not a 14-element first set)**.** The largest cube-difference-free subsets of $\mathbb{Z}/91\mathbb{Z}$ have 14 elements (Lewko's value, reverified by exact search); there are exactly 18200 of them, with only three distinct sets $\Delta R^{(3)}$ among them, and the largest set admissible after any of them has 18 elements. The best period-two chain starting from a 14-element set therefore gives $\gamma = \frac{2}{3} + \frac{3\log 14 + \log 18}{12 \log 91} = 0.8663245\ldots$, which is less than γ_3: in the weighting of Corollary 2.3 the second stage carries one third of the weight of the first, and $\log \frac{28}{18} = 0.4418\ldots$ exceeds $3 \log \frac{14}{13} = 0.2223\ldots$. These three numbers are recomputed by the reproduction script.

4. Exact optimisation over chains

Fix k and a squarefree m such that -1 is a k-th power residue modulo m (automatic for odd k, and true for $k = 2$ when every prime factor of m is $\equiv 1 \pmod 4$). Let P_m be the set of nonzero k-th power residues. For an admissible chain the only information about R_n that constrains R_{n+1} is the set $\Delta R_n^{(k)} \subseteq P_m$, and since ΔR_{n+1} is symmetric under negation only the symmetric closure $F = \Delta R_n^{(k)} \cup (-\Delta R_n^{(k)})$ matters. Call such a symmetric $F \subseteq P_m$ a *state*; there are $2^{|P_m|/2}$ states. For a state F let $\mathcal{R}(F)$ be the family of nonempty $R \subseteq \mathbb{Z}/m\mathbb{Z}$ with $\Delta R \cap F = \emptyset$, i.e. the cliques of the Cayley graph on $\mathbb{Z}/m\mathbb{Z}$ with connection set $(\mathbb{Z}/m\mathbb{Z}) \setminus (F \cup \{0\})$.

4.1. **Definition.** For a state F let $W(F)$ be the supremum of $\sum_{n\geq 1} \log_m |R_n|/k^{n-1}$ over all sequences $(R_n)_{n\geq 1}$ of nonempty subsets of $\mathbb{Z}/m\mathbb{Z}$ with $\Delta R_1 \cap F = \emptyset$ and $\Delta R_{n+1} \cap \Delta R_n^{(k)} = \emptyset$ for all $n \geq 1$.

For $F = P_m$ these sequences, prefixed by $R_0 = \mathbb{Z}/m\mathbb{Z}$, are exactly the k-admissible chains with $R_0 = \mathbb{Z}/m\mathbb{Z}$: condition (3) for $n = 0$ is $\Delta R_1 \cap P_m = \emptyset$, and the side condition $\Delta R_1 \cap \Delta R_1^{(k)} = \emptyset$ follows from it because $\Delta R_1^{(k)} \subseteq P_m$. Hence by (2) the best exponent obtainable from Theorem 2.2 at the modulus m with $R_0 = \mathbb{Z}/m\mathbb{Z}$ is

$$\gamma^*(m) = \frac{k-1}{k} + \frac{k-1}{k^2}\, W(P_m).$$

The function W satisfies the stationary Bellman equation

$$W(F) = \max_{R \in \mathcal{R}(F)} \Big(\log_m |R| + \tfrac{1}{k}\, W\big(\overline{\Delta R^{(k)}}\big) \Big), \tag{4}$$

where the bar denotes symmetric closure. Indeed, the first set of any sequence counted by $W(F)$ is some $R \in \mathcal{R}(F)$ and its tail $(R_n)_{n \geq 2}$ is a sequence counted by $W(\overline{\Delta R^{(k)}})$, since $\Delta R_2 \cap \Delta R^{(k)} = \emptyset$ is equivalent to $\Delta R_2 \cap \overline{\Delta R^{(k)}} = \emptyset$ by symmetry of ΔR_2; this gives $\leq$. Conversely, prepending $R \in \mathcal{R}(F)$ to any sequence counted by $W(\overline{\Delta R^{(k)}})$ gives a sequence counted by $W(F)$; this gives $\geq$. Only $|R|$ and $\overline{\Delta R^{(k)}}$ enter (4), so define, for states F, F',

$$h(F, F') = \max\big\{ |R| : R \in \mathcal{R}(F),\ \Delta R^{(k)} \subseteq F' \big\},$$

a maximum-clique number of the Cayley graph on $\mathbb{Z}/m\mathbb{Z}$ whose connection set is $\{d \neq 0 : d \notin F,\ d^k \in F'\}$. Since W is monotone non-increasing in F (fewer forbidden differences allow more sequences), (4) is equivalent to the finite fixed-point problem

$$W(F) = \max_{F'} \big(\log_m h(F, F') + \tfrac{1}{k} W(F') \big)$$

over the $2^{|P_m|/2}$ states, which, the discount factor $1/k$ being less than 1, has a unique bounded solution, computable by value iteration. As the state space is finite every maximum is attained, and the sequence obtained by following a fixed argmax policy from P_m is eventually periodic; so the supremum defining $W(P_m)$ is attained by an eventually periodic chain.

We solved this program with exact maximum-clique computations for every transition value $h(F, F')$ that the bounds required (a colouring branch-and-bound for graphs of moderate density, and a branch-and-reduce maximum-independent-set solver on the sparse complement for very dense graphs; both are validated on the known values 7 for squares modulo 65, 12 for squares modulo 205 and 14 for cubes modulo 91). Value iteration is run with upper bounds from the relaxation $h(F, F') \leq \min\{h(F, P_m), h(\emptyset, F')\}$ and lower bounds from evaluated transitions, and terminates when they coincide. The following statements are exact.

4.2. **Proposition.** *Let $k = 3$.*

(1) *For $m = 91$ there are 128 states, and $W(P_{91}) = \frac{9}{8}\big(\log_{91} 13 + \frac{1}{3}\log_{91} 28\big)$. Hence $\gamma^*(91) = \gamma_3$: no 3-admissible chain modulo 91, of any length, periodic or not, yields a larger exponent in Theorem 2.2 than the pair of Proposition 3.1, and following the argmax pointers of the solved program from the state P_{91} reproduces that pair.*

(2) *For $m \in \{7, 13, 19, 31, 37, 43\}$ the exact values are*

$$\gamma^*(m) = 0.854858\ldots,\ 0.846825\ldots,\ 0.839444\ldots,\ 0.840590\ldots,\ 0.858737\ldots,\ 0.847023\ldots$$

respectively, and the relaxed upper bounds for $m = 61$ and $m = 67$ are $0.850257\ldots$ and $0.851339\ldots$; all are below γ_3.

Proof. This is the output of the program `dpk.py`, whose transition cache and final bounds are distributed with the paper; the reproduction script re-runs the $m = 91$ case from scratch (it takes about twenty seconds) and re-verifies every cached clique number used in it. □

The scope of these statements is limited to the moduli listed. For other moduli we only have search evidence: penalty-based simulated annealing over periodic chains of periods two and three at $m = 133, 217, 247$ found nothing above 0.858, and the exact program at $m = 133$ (1024 states) was not completed because individual clique computations on 133 vertices at intermediate densities are slow. Larger moduli are penalised by the $1/\log m$ scaling in (2) unless every stage grows in proportion, and for cube-difference-free sets the known upper bounds [7] are consistent with logarithmic growth in the modulus, but we do not claim that 91 is optimal among all moduli.

4.3. **Remark.** The same program applied to $k = 2$, $m = 65$ (1024 states) shows that the four-cycle of sizes $(7, 17, 10, 15)$ of [2] is optimal among all 2-admissible chains modulo 65, with $\gamma^*(65) = 0.773200613\ldots$; the relaxed upper bound alone gives $\gamma^*(65) \le 0.781345\ldots$. We record this for completeness since it was obtained with the same code.

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Appendix A. Verification script

The following Python program, which uses only the standard library, checks Proposition 3.1 and recomputes γ_3.

```
# Standalone verification (Python standard library only) of the certificate for
# sets lacking {x, x+y, x+y^3}: a period-2 admissible chain modulo m = 91.
import math
m, k = 91, 3
A = [0, 31, 36, 41, 46, 51, 56, 61, 66, 71, 76, 81, 86]
B = [0, 7, 9, 11, 13, 20, 22, 24, 26, 33, 35, 37, 39, 46, 48, 50, 52,
     59, 61, 63, 65, 72, 74, 76, 78, 85, 87, 89]
CUBES = {pow(x, k, m) for x in range(m)} - {0}          # nonzero cubes mod 91
diffs = lambda R: {(a - b) % m for a in R for b in R if a != b}
cubim = lambda R: {pow((a - b) % m, k, m) for a in R for b in R if a != b}
assert len(set(A)) == 13 and len(set(B)) == 28
assert all(0 <= x < m for x in A + B)
assert diffs(A) & CUBES == set()      # stage 1: differences of A avoid all nonzero cubes
```

```
assert diffs(B) & cubim(A) == set()   # stage 2: differences of B avoid cubes of differences of
    A
assert diffs(A) & cubim(B) == set()   # wrap-around (automatic, checked anyway)
assert diffs(A) & cubim(A) == set()   # position-zero condition (automatic, checked anyway)
W = (math.log(13) + math.log(28) / 3) / math.log(m) / (1 - 3 ** -2)
gamma = 2 / 3 + (2 / 9) * W
gamma_closed = 2 / 3 + (3 * math.log(13) + math.log(28)) / (12 * math.log(91))
assert abs(gamma - gamma_closed) < 1e-15
print("verified;␣gamma␣=", repr(gamma))
# For comparison: the bound from cube-difference-free sets (Ruzsa/Lewko, m=91, |R|=14)
print("cube-difference␣bound␣(Lewko):", (2 + math.log(14) / math.log(91)) / 3)
```